

ANSWER KEY

Packet D1

Answers verified with SymPy

WRITE THIS CODE ON YOUR PAGE

D1

#	Answer	Skill	Info (bits)	SymPy verification
1	5	Slope of the line through two points	0.38	sympy: $(f(b)-f(a))/(b-a) = 5 = a+b$
2	$\frac{\sqrt{x}+4}{x-16}$	Rationalize using a conjugate	0.44	sympy: $\text{simplify}(\text{original} - \text{key}) == 0$
3	$-\frac{1}{4x}$	Simplify complex fractions (common denominators)	0.69	sympy: $\text{simplify}(\text{expr} - \text{key}) == 0$
4	$x = -6, -5$	Find where a function is undefined (domain)	0.59	sympy: $\text{solve}(\text{den}) = [-6, -5]$
5	$(-4, -2)$	Solve and manipulate absolute-value inequalities	0.69	sympy: $\text{solve_univariate_inequality} = \text{Interval.open}(-4, -2)$
6	$\cos(x) + 1$	Basic trig identities (Pythagorean, quotient)	0.60	sympy: $\text{trigsimp}(\text{expr} - \text{key}) == 0$
7	3	Slopes of secant lines approaching a tangent	0.55	sympy: $(f(q)-f(a))/(q-a) = 3$
8	-24 ft/s	Average velocity approaching instantaneous velocity	0.58	sympy: $(s(b)-s(a))/(b-a) = -24$
9	5	Estimate area under a curve with rectangles	0.65	sympy: $\text{sum } f(i*w)*w = 5$
10	$\frac{19}{12}$	Apply the limit laws	0.43	sympy: substituted value 19/12; denominators nonzero